

The hardest problems will be worked with AI

Provocation · Damian Betebenner, Center for Assessment · AIME-Con 2026, 7 October

One claim

The most important work on the hardest problems in our field will be done with AI.

Not the routine work done faster. The work that was out of reach.

What changed in my own work

	Before AI	With AI
Theory	months per derivation, one person	a derivation tested against a collaborator in an afternoon
Code	the bottleneck	written, refactored, and tested alongside the theory
Simulation	a few designs, chosen for feasibility	thousands of conditions, chosen for the question
Documentation	written after, if at all	written with the work
Ambition	sized to what one person could check	sized to the problem

Scale, speed, and *ambition* change. So does the demand for verification and judgment.

One example: an advanced mathematical task

1. **Conjecture:** a property of an advanced measurement model [name the problem]
2. **Derivation:** worked through with AI as an interlocutor; each step challenged
3. **Implementation:** the derivation becomes code the same week
4. **Simulation at scale:** the result stress-tested across thousands of conditions
5. **Verification:** what I could check, what I could not

Before: a multi-year research program for one methodologist.

With AI: weeks.

What did not change: I answer for the result.

One unresolved problem

When the derivation and the simulation exceed what the author, or any reviewer, can check by hand, what makes the result warranted, and who answers for it?

Guiding questions 3 and 5. See also Rijmen: *owning results one did not produce by hand*; Briggs: *whose judgment is it?*